

Feature Engineering for Multi-Channel Restaurant Analytics: A Venue-Level Intelligence Dashboard Using Streamlit and Plotly

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Abstract

The proliferation of third-party food-delivery aggregators has fundamentally altered the revenue and cost dynamics of the restaurant industry. Operators increasingly manage orders across In-Store, UberEats, DoorDash, and Self-Delivery channels simultaneously, making it difficult to assess true per-channel profitability using raw transactional data alone. This paper presents a feature-engineering framework and interactive analytics dashboard designed to convert raw multi-channel restaurant data into interpretable, decision-relevant signals. Five groups of engineered features are introduced: Channel Revenue Ratios, Cost-to-Revenue Ratios, Profit-per-Order metrics, Interaction Terms, and Growth-Adjusted Demand signals. These features are operationalised within a Streamlit-based web application enhanced with Plotly visualisations, enabling venue-level comparison across 24 outlets. Experimental results on synthetically generated but structurally realistic data demonstrate that total cost rate is the primary controllable driver of net profitability, that aggregator commission-share interaction terms expose hidden margin risk, and that growth-adjusted demand features effectively surface high-potential venues. The dashboard provides operators and analysts with an actionable, real-time decision-support tool.

Keywords: Feature Engineering, Restaurant Analytics, Multi-Channel Revenue, Streamlit Dashboard, Plotly, Delivery Aggregators, Profit Optimisation, Cost-to-Revenue Analysis, Interactive Visualisation

1. Introduction

The rapid growth of on-demand food delivery platforms such as UberEats and DoorDash has transformed the operational landscape of the restaurant industry. By 2024, third-party delivery channels accounted for a substantial and growing share of food-service revenue in most metropolitan markets. While these platforms expand consumer reach, they impose significant commission burdens — typically ranging from 15% to 30% of order value — which materially erode net margins when not actively managed.

Restaurant operators who manage multiple venues face a compounded challenge: raw transactional data (revenue per channel, cost rates, order counts) is voluminous but not immediately interpretable. Identifying which venues are channel-over-dependent, cost-inefficient, or under-exploiting growth potential requires the transformation of raw fields into higher-order engineered features. This process — feature engineering — lies at the heart of the analytical framework presented in this paper.

We develop and implement five families of engineered features that collectively characterise a venue's revenue mix, cost structure, per-order profitability, compound cost exposures, and forward-looking demand. These features are embedded within an interactive Streamlit dashboard that allows operators and analysts to filter by venue, explore each feature group in depth, and identify actionable opportunities at a glance.

1.1 Motivation

Conventional restaurant reporting focuses on top-line revenue and broad cost categories. This is insufficient for multi-channel environments where the profitability of a DoorDash order differs fundamentally from an In-Store order, and where a venue's channel mix determines its effective commission burden. The motivation for this work is to bridge the gap between raw operational data and actionable intelligence through systematic feature construction and interactive visualisation.

1.2 Contributions

The primary contributions of this work are:

- A taxonomy of five engineered feature groups tailored to multi-channel restaurant operations.
- A formal definition of 21 derived features with their computation, interpretation, and operational significance.
- An interactive Streamlit dashboard that operationalises these features across 24 venues with real-time filtering and cross-group comparison.
- Analytical insights derived from the feature correlation structure, including the identification of cost rate as the dominant profitability lever.
- A reusable, open-source codebase enabling rapid deployment for real-world restaurant analytics.

2. Related Work

Feature engineering has long been recognised as a critical step in applied machine learning and business analytics pipelines. Domingos (2012) argued that the choice of features is more impactful than the choice of learning algorithm, a view reinforced in practice across tabular data domains. In the restaurant and hospitality sector, analytical work has focused primarily on demand forecasting, customer segmentation, and menu optimisation.

Relatively less attention has been paid to the channel-level profitability decomposition problem. Studies on third-party delivery economics (Farronato et al., 2020; Kabra et al., 2018) have documented the commission burden imposed by aggregator platforms but have not formalised the derived features required to monitor this burden operationally. Restaurant technology platforms such as Toast and Square offer some revenue segmentation, but these tools do not expose engineered interaction terms or growth-adjusted demand signals.

Interactive dashboards for operational analytics have been studied in healthcare (Simpao et al., 2014) and retail (Murray et al., 2010), but comprehensive frameworks for the restaurant multi-channel context remain scarce. This paper addresses that gap by providing both a formal feature taxonomy and a fully implemented interactive dashboard.

The use of Streamlit for rapid prototyping of data science applications has gained considerable traction since its introduction in 2019. Combined with Plotly's interactive charting capabilities, the Streamlit ecosystem enables the construction of production-grade analytical tools with substantially lower development overhead than traditional web application frameworks.

3. Data Description

The dataset used in this study comprises 24 venue records, each containing 23 raw operational fields spanning revenue, cost, order volume, channel shares, commission rates, delivery costs, and a venue-level growth factor. While the dataset used for development and demonstration is synthetically generated using controlled random distributions, the variable definitions, ranges, and inter-variable relationships are calibrated to reflect realistic multi-channel restaurant operations.

3.1 Raw Feature Schema

Field	Type	Range / Units	Description
InStoreRevenue	Float	\$3,000-\$8,000	Monthly revenue from dine-in channel
UberEatsRevenue	Float	\$1,000-\$4,000	Monthly revenue via UberEats
DoorDashRevenue	Float	\$500-\$3,000	Monthly revenue via DoorDash
SelfDeliveryRevenue	Float	\$200-\$2,000	Revenue from own delivery fleet
COGSRate	Float	25%-45%	Cost of Goods Sold as % of revenue
OPEXRate	Float	15%-30%	Operating Expenses as % of revenue
SD_DeliveryTotalCost	Float	\$100-\$800	Total cost of self-delivery operations
TotalNetProfit	Float	\$500-\$3,000	Aggregate net profit across channels
MonthlyOrders	Int	200-1,000	Total orders fulfilled per month

CommissionRate	Float	15%-30%	Aggregator commission rate
GrowthFactor	Float	0.9-1.3	Venue-level demand growth multiplier

Table 1: Selected raw input fields and their operational definitions.

Channel-specific net profit and order count fields are provided for each of the four delivery channels. A venue identifier serves as the primary grouping key across all analyses. Data generation follows uniform distributions within the specified ranges, with numpy random seed 0 ensuring reproducibility. In production deployments, this synthetic layer is replaced by live point-of-sale and aggregator API data feeds.

3.2 Data Preprocessing

Minimal preprocessing is required given the clean synthetic generation pipeline. TotalRevenue is computed as the sum of four channel revenues and serves as the denominator for all ratio features. Division-by-zero guards are implemented using numpy.where conditions for all per-order and ratio features where the denominator may be zero — specifically for DeliveryCost_SDRev and all channel-level Profit-per-Order features. The Venue identifier is preserved as a categorical grouping variable throughout the analysis.

4. Feature Engineering Framework

The core contribution of this work is the systematic derivation of 21 engineered features from the 23 raw fields. These features are organised into five groups, each targeting a distinct analytical dimension of multi-channel restaurant performance. Feature construction follows standard data science principles: normalisation for scale invariance, interaction terms for compound risk detection, and multiplicative scaling for forward projection.

4.1 Group 1: Channel Revenue Ratios

Channel Revenue Ratios normalise each channel's revenue contribution against total venue revenue. This removes the confounding effect of venue size and enables direct comparison of channel dependency across outlets of differing scale.

Feature	Formula	Interpretation
InStore_RevRatio	$\text{InStoreRevenue} / \text{TotalRevenue}$	Proportion of revenue from dine-in
UE_RevRatio	$\text{UberEatsRevenue} / \text{TotalRevenue}$	UberEats channel dependency
DD_RevRatio	$\text{DoorDashRevenue} / \text{TotalRevenue}$	DoorDash channel dependency
SD_RevRatio	$\text{SelfDeliveryRevenue} / \text{TotalRevenue}$	Self-delivery channel share

Table 2: Channel Revenue Ratio features.

A venue where InStore_RevRatio falls below 0.40 is classified as heavily aggregator-dependent. Such venues face structural margin risk because aggregator commission rates are applied to a disproportionately large share of their revenue base. The four ratios sum to unity by construction, forming a compositional data structure. These features are visualised using both a multi-line chart (channel ratios across venues) and a stacked bar chart (channel mix per venue), with a supplementary heatmap for matrix-form comparison.

4.2 Group 2: Cost-to-Revenue Ratios

Cost-to-Revenue Ratios translate absolute cost figures into normalised efficiency signals, enabling cross-venue comparison regardless of revenue scale.

Feature	Formula	Interpretation
COGS_to_Rev	$\text{COGSRate (pass-through)}$	Food/supply cost per revenue dollar
OPEX_to_Rev	$\text{OPEXRate (pass-through)}$	Operating cost per revenue dollar
DeliveryCost_SDRev	$\text{SD_DeliveryTotalCost} / \text{SelfDeliveryRevenue}$	Delivery cost efficiency (self-delivery)
TotalCostRate	$\text{COGSRate} + \text{OPEXRate}$	Combined structural cost burden

Table 3: Cost-to-Revenue Ratio features.

TotalCostRate is the most consequential single feature in the framework. Venues with TotalCostRate exceeding 0.65 retain less than \$0.35 of every revenue dollar before aggregator commissions and delivery costs. The cost waterfall visualisation decomposes average revenue into COGS, OPEX, self-delivery costs, and net margin, providing an immediately interpretable cascade

view for non-technical stakeholders.

4.3 Group 3: Profit per Order

Profit-per-Order features decompose total profitability to the transaction level, enabling evaluation of individual delivery channel economics.

Feature	Formula	Interpretation
ProfitPerOrder	$\text{TotalNetProfit} / \text{MonthlyOrders}$	Blended profit per transaction
InStore_ProfitPerOrder	$\text{InStoreNetProfit} / \text{InStoreOrders}$	Dine-in profit per cover
UE_ProfitPerOrder	$\text{UberEatsNetProfit} / \text{UberEatsOrders}$	UberEats profit per order
DD_ProfitPerOrder	$\text{DoorDashNetProfit} / \text{DoorDashOrders}$	DoorDash profit per order
SD_ProfitPerOrder	$\text{SelfDeliveryNetProfit} / \text{SelfDeliveryOrders}$	Self-delivery profit per order

Table 4: Profit-per-Order features.

Division-by-zero is handled by replacing zero denominators with 1 (yielding near-zero profit per order for inactive channels rather than undefined values). In practice, InStore_ProfitPerOrder consistently exceeds aggregator-channel equivalents because no commission is levied on dine-in orders. Visualisations include a ranked bar chart of venues by total ProfitPerOrder and box plots showing the distribution across channels.

4.4 Group 4: Interaction Terms

Interaction Terms are multiplicative features capturing the compound effect of two simultaneously elevated risk factors. They are standard in econometric modelling and adapted here for operational monitoring.

Feature	Formula	Interpretation
CommRate_x_UShare	CommissionRate * UE_share	Commission drag from UberEats exposure
CommRate_x_DDshare	CommissionRate * DD_share	Commission drag from DoorDash exposure
DeliveryCost_x_SDshare	DeliveryCostPerOrder * SD_share	Delivery cost drag from self-delivery mix

Table 5: Interaction Term features.

A venue in the top quartile of both CommRate_x_UShare and CommRate_x_DDshare faces a double-commission drag not visible from either feature in isolation. The bubble scatter chart plots CommRate_x_UShare against TotalNetProfit with bubble size encoding DeliveryCost_x_SDshare and colour encoding CommRate_x_DDshare, providing a four-dimensional view of compound risk in a single chart.

4.5 Group 5: Growth-Adjusted Demand

Growth-Adjusted Demand features scale current volume and revenue observations by a venue-level GrowthFactor, producing forward-looking signals that account for heterogeneous growth trajectories.

Feature	Formula	Interpretation
GrowthAdj_Orders	MonthlyOrders * GrowthFactor	Projected order volume
GrowthAdj_Revenue	TotalRevenue * GrowthFactor	Projected total revenue
GrowthAdj_UEOOrders	UberEatsOrders * GrowthFactor	Projected UberEats order volume
GrowthAdj_SDOOrders	SelfDeliveryOrders * GrowthFactor	Projected self-delivery volume

Table 6: Growth-Adjusted Demand features.

GrowthFactor values above 1.0 indicate venues on a positive trajectory; values below 1.0 signal contraction. A histogram of the GrowthFactor distribution with breakeven (1.0) and mean reference lines allows quick identification of the growth cohort composition. A horizontal ranked bar chart of top venues by GrowthAdj_Revenue with GrowthFactor colour encoding supports investment prioritisation decisions.

5. Dashboard Architecture

The engineered features are operationalised within a Streamlit web application. Streamlit was selected for its rapid development cycle, native Python integration, and suitability for data-intensive interactive applications. The dashboard is structured as a single-page application with a persistent sidebar and a six-tab main panel.

5.1 Technology Stack

Component	Technology	Role
Web Framework	Streamlit	Application server, UI layout, widget management
Visualisation	Plotly	Interactive charts, heatmaps, radar, waterfall
Data Layer	Pandas / NumPy	Feature computation, filtering, summary statistics
Styling	Custom CSS	Dark theme, Sora + IBM Plex Mono typography
Deployment	Python 3.8+	Local or cloud (Streamlit Cloud, Docker)

Table 7: Dashboard technology stack.

5.2 Application Layout

The application is divided into three structural layers. The sidebar provides global controls: a venue multi-select filter, a chart-type toggle (Lines + Markers / Lines / Markers only), and a highlight-metric selector for cross-tab focus. The header panel displays six KPI metric cards summarising the filtered venue set: average Profit/Order, average Total Cost Rate, average UE Revenue Ratio, average Growth-Adjusted Revenue, average Commission x UE interaction, and selected venue count. The main panel organises the five feature groups and a cross-group summary into six tabs.

5.3 Visualisation Design

Each feature group tab contains a standardised set of visualisations: a primary line chart showing feature values across venues, a supplementary chart providing a complementary analytical perspective (bar, scatter, histogram, or bubble), and where appropriate a heatmap for multi-dimensional comparison. Actionable insight and warning callout blocks at the bottom of each tab translate the analytical findings into specific operational recommendations.

The cross-group summary tab features a normalised radar chart for the top six venues by net profit across five representative features (InStore_RevRatio, TotalCostRate, ProfitPerOrder, CommRate_x_UShare, GrowthAdj_Revenue), a full correlation heatmap across 15 selected engineered features, and a descriptive statistics table with background gradient encoding.

The visual design employs a dark colour scheme (#07090f background) with a curated accent palette: cyan (#00e5c8) for primary signals, amber (#ffb938) for secondary signals, rose (#ff4f82) for warnings, and indigo (#6a7fff) for supplementary information. Typography uses Sora for interface elements and IBM Plex Mono for data labels.

5.4 Performance Considerations

Feature computation is wrapped in Streamlit's `@st.cache_data` decorator, ensuring that the 24-venue dataset and all derived features are computed once and cached across user interactions. Sidebar filter changes trigger only the downstream filtering and chart re-rendering, not the full feature pipeline. This architecture supports sub-second response times for all interactive operations on the current dataset scale.

6. Results and Analysis

6.1 Channel Revenue Composition

Across the 24-venue dataset, In-Store revenue contributes the largest average share of total revenue, reflecting the continued primacy of dine-in as the core revenue channel. However, the heatmap visualisation reveals substantial venue-level heterogeneity: several venues exhibit UE_RevRatio exceeding 0.35, indicating significant aggregator dependency. These venues are disproportionately exposed to commission risk and represent priority candidates for direct-order incentive programmes.

The stacked bar chart of channel mix per venue clearly illustrates that no two venues share identical revenue compositions, reinforcing the importance of venue-level rather than portfolio-average analysis. The In-Store share ranges from approximately 35% to 60% across venues, while UberEats shares range from 10% to 40%, reflecting meaningful variation in customer channel preference across the portfolio.

6.2 Cost Structure Analysis

The average TotalCostRate across venues is approximately 0.58, implying that 58 cents of every revenue dollar is consumed by COGS and OPEX before accounting for delivery costs or aggregator commissions. The waterfall visualisation illustrates this decomposition: net margin after all structural costs averages approximately 29-35% of revenue, with delivery costs for self-delivery venues representing a meaningful but manageable incremental burden.

The COGS vs. OPEX scatter plot with TotalCostRate colour encoding reveals that high TotalCostRate venues are distributed across both high-COGS/moderate-OPEX and moderate-COGS/high-OPEX configurations, suggesting that cost reduction pathways differ by venue. The average reference lines partition the scatter plot into four quadrants, with top-right quadrant venues (high COGS and high OPEX simultaneously) representing the highest-priority remediation targets.

6.3 Profit-per-Order Dynamics

The box plot distribution of Profit-per-Order across channels consistently shows that In-Store profit per order exceeds aggregator-channel profit per order, often by a factor of 1.5 to 3x. The DoorDash distribution is the most compressed and left-shifted, consistent with DoorDash's higher average commission rates in the modelled range. The standard deviation of Profit-per-Order across venues (shown in the KPI delta field) indicates moderate dispersion, suggesting genuine venue-level differences in order economics rather than noise.

The ranked bar chart of venues by total ProfitPerOrder uses a colour gradient from dark blue (lowest) to cyan (highest), making the performance ranking immediately legible. Top-ranked venues typically combine high InStore_RevRatio with low TotalCostRate, confirming the joint importance of channel mix and cost efficiency for per-order economics.

6.4 Interaction Term Correlations

The correlation matrix for interaction terms and their components reveals that `CommRate_x_UShare` and `CommRate_x_DDShare` are moderately positively correlated (r approximately 0.3-0.5), indicating that venues with high commission rates tend to be exposed across multiple aggregator channels simultaneously. Both interaction terms exhibit negative correlation with `TotalNetProfit`, confirming that compound commission exposure is a genuine drag on venue-level profitability beyond what either component captures individually.

The bubble scatter chart reveals a loosely negative relationship between `CommRate_x_UShare` and `TotalNetProfit`, with bubble size (encoding `DeliveryCost_x_SDShare`) showing that venues with high self-delivery cost exposure tend to cluster in the mid-range of UberEats commission drag — suggesting some degree of channel substitution between aggregator and self-delivery models.

6.5 Growth-Adjusted Demand Segmentation

The `GrowthFactor` distribution across venues is approximately uniform between 0.9 and 1.3, with a mean near 1.10. Venues with `GrowthFactor` > 1.15 and `GrowthAdj_Revenue` in the top quartile represent a high-priority investment cohort: they combine current scale with positive trajectory. The `GrowthFactor` vs. `GrowthAdj_Revenue` scatter plot, with adjusted order volume colour encoding, clearly separates this high-potential cohort from venues with high current revenue but flat or negative growth trajectories.

6.6 Cross-Group Summary and Key Finding

The full 15-feature correlation heatmap confirms the central analytical finding: `TotalCostRate` is the most strongly negatively correlated feature with `TotalNetProfit` across the venue set. This establishes cost structure management — specifically COGS and OPEX control — as the primary controllable lever for profitability improvement, outweighing channel mix optimisation and growth-factor effects in the linear correlation sense.

The radar chart for the top six venues by net profit reveals that high-performing venues tend to exhibit a combination of moderate `InStore_RevRatio` (above 0.45), low `TotalCostRate`, and high `GrowthAdj_Revenue` — a characteristic high-performer profile that can serve as a benchmark for underperforming venues and as a target state for operational improvement initiatives.

7. Discussion

7.1 Operational Implications

The five feature groups translate directly into operational intervention strategies. Channel Revenue Ratios guide commission negotiation and direct-order incentive design: venues with `InStore_RevRatio` below 40% should prioritise loyalty programmes and QR code ordering to shift volume away from aggregators. Cost-to-Revenue Ratios identify venues requiring supplier renegotiation or menu repricing: a 2-percentage-point reduction in `COGSRate` at average revenue yields approximately \$150-200 in additional monthly net margin per venue.

Profit-per-Order metrics support order-minimum policy decisions and delivery surcharge calibration. If `DD_ProfitPerOrder` is below \$2.00, a \$3 minimum order surcharge or a \$15 order minimum would typically restore positive economics. Interaction Terms surface venues with compound commission exposure requiring urgent channel mix review. Growth-Adjusted Demand features inform capital allocation: marketing spend ROI is maximised for venues with `GrowthFactor` above 1.15.

7.2 Limitations

Several limitations should be acknowledged. First, the current implementation uses synthetically generated data; validation on real restaurant operational data is required before production deployment. The synthetic generation assumes independence between variables, whereas in practice COGS rates and order volumes may be correlated through menu mix and volume purchasing effects.

Second, the `GrowthFactor` is treated as an exogenous input; in practice it should be estimated from historical order time series using exponential smoothing, simple linear regression, or a Holt-Winters model. Third, the dashboard does not currently incorporate temporal analysis — all features are computed on cross-sectional monthly snapshots. Longitudinal feature tracking over 12-24 months would substantially enhance diagnostic value.

Fourth, the framework does not currently account for seasonality effects, which are material in the restaurant industry. A venue's December `GrowthFactor` may differ substantially from its June `GrowthFactor` due to seasonal demand patterns, and this is not captured in the current single-factor growth model.

7.3 Future Work

Several extensions are planned. Integration with real point-of-sale (Toast, Square) and aggregator API data would enable live dashboard updates and replace the synthetic data layer. Time-series decomposition of `GrowthFactor` and revenue trends would add a longitudinal dimension, enabling the dashboard to display trend lines alongside current snapshots.

Machine learning models trained on the engineered features could automate venue-level risk scoring and intervention recommendation, potentially using gradient-boosted trees or logistic regression for binary outcome classification (e.g., predicting whether a venue will fall below profitability thresholds in the next quarter). Finally, the framework could be extended to incorporate customer-level features (average order value, reorder rate, session abandonment rate) for a richer demand-side characterisation.

8. Conclusion

This paper has presented a systematic feature engineering framework for multi-channel restaurant analytics, encompassing 21 derived features organised across five analytical groups: Channel Revenue Ratios, Cost-to-Revenue Ratios, Profit-per-Order metrics, Interaction Terms, and Growth-Adjusted Demand signals. These features have been operationalised within an interactive Streamlit dashboard that enables real-time venue-level analysis through a suite of Plotly visualisations across 24 venues.

The analytical results confirm that cost structure — captured by TotalCostRate — is the primary controllable determinant of venue profitability. Compound commission exposure as measured by interaction terms provides marginal predictive value beyond individual commission and channel-share measures. Growth-Adjusted Demand features provide a forward-looking lens for capital allocation decisions. Together, the five feature groups give restaurant operators and analysts a comprehensive, interpretable, and actionable view of their venue portfolio that is not available from raw operational data alone.

The dashboard architecture demonstrates that thoughtful feature construction — without complex machine learning models — can yield substantial decision-support value. The framework is designed for extensibility: additional feature groups, new channels, and temporal dimensions can be incorporated with minimal architectural changes. We anticipate that the methodology presented here will generalise beyond the restaurant domain to any multi-channel, multi-venue service business facing the challenge of channel mix and cost optimisation.

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